

Unit 1

Cardinal Numbers

Equivalent Sets

(1) $f: A \rightarrow B$ is one - one $\Leftrightarrow$ Let $f(x) = f(y) \Rightarrow x = y, \forall x, y \in A$ (or) $x \neq y \Rightarrow f(x) \neq f(y), \forall x, y \in A$ (2) $f: A \rightarrow B$ is onto $\Leftrightarrow$ $\forall y \in B, \exists x \in A$ such that $f(x) = y$ (3) $A \sim B \Leftrightarrow$ a function $f: A \rightarrow B$ is

one - one and onto

1. Prove that $[0, 1] \sim (0, 1)$.Proof: $[0, 1] = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup A$

$$(0, 1) = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} \cup A$$

where $A = [0, 1] - \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$

$$A = (0, 1) - \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$$

Let $f: [0, 1] \rightarrow (0, 1)$ be defined by

$$f(x) = \begin{cases} 0, & x = 0 \\ \frac{1}{n+2}, & x = \frac{1}{n}, \forall n \in \mathbb{N} \\ x, & x \neq 0, \frac{1}{n}, x \in A \end{cases}$$

Clearly, f is one - one.Take any $y = \frac{1}{2} \in (0, 1), \exists x = 0 \in [0, 1] \Rightarrow f(x) = f(0) = \frac{1}{2} = y$ Take any $y \in \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}, \exists x = \frac{1}{n} = \frac{y}{1-2y} \in [0, 1]$ such that

$$f(x) = f\left(\frac{1}{n}\right) = \frac{1}{n+2} = \frac{1}{\frac{1}{1-2y} + 2} = \frac{1}{\frac{1+2-4y}{1-2y}} = \frac{1-2y}{1-2y+2y} = y$$

 $\therefore f$ is onto.Since f is one-one and onto, $[0, 1] \sim (0, 1)$.2. Prove that $[0, 1] \sim [0, 1)$.Proof: $[0, 1] = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup A$

$$[0, 1) = \{0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} \cup A$$

where $A = [0, 1] - \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$

$$A = [0, 1) - \{0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$$

Let $f: [0, 1] \rightarrow [0, 1)$ be defined by

$$f(x) = \begin{cases} 0, & x = 0 \\ \frac{1}{n+1}, & x = \frac{1}{n}, \forall n \in \mathbb{N} \\ x, & x \neq 0, \frac{1}{n}, x \in A \end{cases}$$

Clearly, f is one - one.Take $y = 0 \in [0, 1), \exists x = 0 \in [0, 1] \Rightarrow f(x) = f(0) = 0 = y$ Take any $y \in \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}, \exists x = \frac{1}{n} = \frac{y}{1-y} \in [0, 1]$ such that

$$\begin{aligned} f(x) &= f\left(\frac{1}{n}\right) = f\left(\frac{y}{1-y}\right) = f\left(\frac{1}{\frac{1-y}{y}}\right) \\ &= \frac{1}{\frac{1-y}{y} + 1} = \frac{y}{1-y+y} = y \end{aligned}$$

 $\therefore f$ is onto.Since f is one-one and onto, $[0, 1] \sim [0, 1)$.3. Prove that $[0, 1] \sim (0, 1)$.Proof: $[0, 1] = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup A$

$$(0, 1) = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} \cup A$$

where $A = [0, 1] - \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$

$$A = (0, 1) - \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$$

Let $f: [0,1] \rightarrow (0,1]$ be defined by

$$f(x) = \begin{cases} 1, & x = 0 \\ \frac{1}{n+1}, & x = \frac{1}{n}, \forall n \in \mathbb{N} \\ x, & x \neq 0, \frac{1}{n}, x \in A \end{cases}$$

Clearly, f is one-one.

Take any $y = 1 \in (0,1], \exists x = 0 \in [0,1] \ni f(x) = f(0) = 1 = y$

Take any $y \in \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}, \exists x = \frac{1}{n} = \frac{y}{1-y} \in [0,1]$

$$\begin{aligned} \text{such that } f(x) &= f\left(\frac{1}{n}\right) = \frac{1}{n+1} \\ &= \frac{1}{\frac{1-y}{1-y} + 1} = \frac{y}{1-y+y} = y \end{aligned}$$

$\therefore f$ is onto.

Since f is one-one and onto, $[0,1] \sim (0,1]$.

Product Sets

$$A \times B = \{(a,b) \mid a \in A, b \in B\}$$

$$A \times B \times C = \{(a,b,c) \mid a \in A, b \in B, c \in C\}$$

$$(A \times B) \times C = \{((a,b),c) \mid (a,b) \in A \times B, a \in A, b \in B, c \in C\}$$

$$A \times (B \times C) = \{(a,(b,c)) \mid a \in A, (b,c) \in B \times C, b \in B, c \in C\}$$

4. Prove that $A \times (B \times C) \sim A \times B \times C$ for any sets A, B and C .

Solution: Let $f: A \times (B \times C) \rightarrow A \times B \times C$ be a function defined by

$$f(a,(b,c)) = (a,b,c), \forall a \in A, b \in B, c \in C, (b,c) \in B \times C$$

Take any $x, y \in A \times (B \times C)$

$$x = (a,(b,c)), y = (d,(e,f))$$

where $(b,c), (e,f) \in B \times C, \forall b, e \in B, c, f \in C, a, d \in A$

$$\text{Let } f(x) = f(y) \quad \forall x, y \in A \times (B \times C)$$

$$f(a,(b,c)) = f(d,(e,f))$$

$$(a,b,c) = (d,e,f)$$

$$a = d, b = e, c = f$$

$$(a,(b,c)) = (d,(e,f))$$

$$x = y$$

$\therefore f$ is one-one.

Take any

$$y = (a,b,c) \in A \times B \times C,$$

$\exists x = (a,(b,c)) \in A \times (B \times C)$ such that

$$f(x) = f(a,(b,c)) = (a,b,c) = y$$

$\therefore f$ is onto.

Since f is one-one and onto,

$$A \times (B \times C) \sim A \times B \times C.$$

5. Prove that $(A \times B) \times C \sim A \times B \times C$ for any sets A, B and C .

Solution: Let $f: (A \times B) \times C \rightarrow A \times B \times C$ be a function defined by

$$f((a,b),c) = (a,b,c), \forall a \in A, b \in B, c \in C, (a,b) \in A \times B$$

Take any $x, y \in (A \times B) \times C$

$$x = ((a,b),c), y = ((d,e),f)$$

where $(a,b), (d,e) \in A \times B, \forall a, d \in A, b, e \in B, c, f \in C$

$$\text{Let } f(x) = f(y) \quad \forall x, y \in (A \times B) \times C$$

$$f((a,b),c) = f((d,e),f)$$

$$(a,b,c) = (d,e,f)$$

$$\therefore a = d, b = e, c = f$$

$$(a,b) = (d,e)$$

$$\therefore ((a,b),c) = ((d,e),f)$$

$$x = y$$

$\therefore f$ is one-one.

Take any

$$y = (a,b,c) \in A \times B \times C,$$

$\exists x = ((a,b),c) \in (A \times B) \times C$ such that

$$f(x) = f((a,b),c) = (a,b,c) = y$$

$\therefore f$ is onto.

Since f is one-one and onto,

$$(A \times B) \times C \sim A \times B \times C.$$

6. Prove that $Z \sim N$.

Proof:

Let Z = the set of all integers.

$$= \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

Let N = the set of all natural numbers

$$= \{1, 2, 3, \dots\}$$

Consider the function $f: N \rightarrow Z$ defined by

$$f(x) = \begin{cases} \frac{x}{2}, & \text{if } x \text{ is even} \\ \frac{1-x}{2}, & \text{if } x \text{ is odd.} \end{cases}$$

It is clear that f is one-one.

Take any $y \in Z$,

$$\exists x = 2y \in N \ni f(x) = \frac{x}{2} = \frac{2y}{2} = y$$

Take any $y \in Z$, $\exists x = 1 - 2y \in N$ such that

$$f(x) = \frac{1-x}{2} = \frac{1-(1-2y)}{2} = y$$

$\therefore f$ is onto.

Since f is both 1-1 and onto, $Z \sim N$.

7. Let A and B be any two sets. Prove that

$$A \sim A \times \{1\} \text{ and } B \sim B \times \{2\}$$

Proof: Let $f: A \rightarrow A \times \{1\}$ be defined by

$$f(x) = (x, 1), \forall x \in A$$

$$\text{Let } f(x) = f(y), \forall x, y \in A$$

$$(x, 1) = (y, 1)$$

$$x = y$$

$\therefore f$ is one-one.

Take any $y = (a, 1) \in A \times \{1\}$, $\exists x = a \in A$

such that $f(x) = (x, 1) = (a, 1) = y$.

Since f is one-one and onto, $A \sim A \times \{1\}$.

Let $g: B \rightarrow B \times \{2\}$ be defined by

$$g(x) = (x, 2), \forall x \in B$$

Let $g(x) = g(y)$, $x, y \in B$

$$(x, 2) = (y, 2)$$

$$x = y$$

Therefore g is one-one.

Take any $y = (b, 2) \in B \times \{2\}$, $\exists x = b \in B$

such that $g(x) = (x, 2) = (b, 2) = y$

$\therefore g$ is onto.

Since g is one-one and onto, $B \sim B \times \{2\}$.

Denumerable

A is denumerable $\Leftrightarrow$

$$(1) A \sim N$$

(2) $A = \{x_1, x_2, \dots\}$ is an infinite sequence of distinct elements.

8. Let $M = \{0, 1, 2, \dots\} = N \cup \{0\}$ for each natural number $n \in N$ can be written uniquely in the form

$n = 2^s(2r+1)$, where $r, s \in M$ Consider the

function $f: N \rightarrow M \times M$ defined by

$f(n) = (s, r)$ where r and s are as above. Prove that

$M \times M$ is denumerable.

Proof: Let $f: N \rightarrow M \times M$ be defined by

$$f(n) = (s, r), \forall s, r \in M, n \in N.$$

Let $n, m \in N$

$$n = 2^s(2r+1), m = 2^p(2q+1), \forall r, s, p, q \in M$$

Let $f(n) = f(m)$, $\forall n, m \in N$

$$(s, r) = (p, q)$$

$$s = p, r = q$$

$$2^s(2r+1) = 2^p(2q+1)$$

$$n = m$$

So f is one-one.

Take any

$y = (s, r) \in M \times M$, $\exists n \in N \ni f(n) = (s, r) = y$

So f is onto.

Since f is one-one and onto, $N \sim M \times M$

$\therefore M \times M$ is denumerable.

9. Prove that the set of rational numbers Q is denumerable.

Proof: Let Q^+ be the set of positive rational numbers and Q^- be the set of negative rational numbers.

Then $Q = Q^- \cup \{0\} \cup Q^+$ is the set of rational numbers.

Let $f: Q^+ \rightarrow N \times N$ be defined by

$$f\left(\frac{p}{q}\right) = (p, q), \quad \frac{p}{q} \in Q^+, \quad p, q \in \mathbb{Z}^+.$$

Let $f\left(\frac{p}{q}\right) = f\left(\frac{r}{s}\right), \frac{p}{q}, \frac{r}{s} \in Q^+$

$$(p, q) = (r, s)$$

$$\therefore p = r, q = s$$

$$\frac{p}{q} = \frac{r}{s}$$

$\therefore f$ is one-one.

$\therefore Q^+$ is equivalent to a subset of $N \times N$.

Since $N \times N$ is denumerable, then a subset of $N \times N$ is denumerable.

$\therefore Q^+$ is denumerable.

Similarly, Q^- is denumerable.

$\therefore Q = Q^- \cup \{0\} \cup Q^+$ is denumerable.

$\therefore Q$ is denumerable.

10. Prove that Z , the set of all integers is denumerable.

Proof: Let Z = the set of all integers.

Let $f: N \rightarrow Z$ be defined as follows.

$$N: 1 \quad 2 \quad 3 \quad 4 \quad 5 \dots$$

$$f \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$Z: 0 \quad 1 \quad -1 \quad 2 \quad -2 \dots$$

There is a one-one, onto function f from N to Z defined by

$$f(n) = \begin{cases} \frac{n}{2}, & n \text{ is even} \\ -\frac{(n-1)}{2}, & n \text{ is odd.} \end{cases}$$

Thus $N \sim Z$ and so $Z \sim N$.

So Z is denumerable.

11. Prove that $2N$, the set of even integers is denumerable.

Proof: Let $2N$ = the set of even integers.

Let $f: N \rightarrow 2N$ be defined as follows.

$$N: 1 \quad 2 \quad 3 \quad 4 \quad 5 \dots$$

$$f \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$2N: 2 \quad -2 \quad 4 \quad -4 \quad 6 \dots$$

There is a one-one, onto function f from N to $2N$ defined by

$$f(n) = \begin{cases} n+1, & n \text{ is odd} \\ -n, & n \text{ is even.} \end{cases}$$

Thus $N \sim 2N$ and so $2N \sim N$.

So $2N$ is denumerable.

12. Prove that the product set $N \times N$ is denumerable.

Solution: Consider the product set $N \times N$.

$$(1,1) \quad (1,2) \rightarrow (1,3) \quad (1,4) \rightarrow \dots$$

$$\downarrow \quad \nearrow \quad \swarrow \quad \nearrow \quad \swarrow$$

$$(2,1) \quad (2,2) \quad (2,3) \quad (2,4) \quad \dots$$

$$\swarrow \quad \nearrow \quad \swarrow \quad \nearrow$$

$$(3,1) \quad (3,2) \quad (3,3) \quad (3,4) \quad \dots$$

$$\downarrow \quad \nearrow \quad \swarrow \quad \nearrow \quad \swarrow$$

$$(4,1) \quad (4,2) \quad (4,3) \quad (4,4) \quad \dots$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \dots$$

The product set $N \times N$ can be written in an infinite sequence of distinct elements as follow,

$$\{(1,1), (2,1), (1,2), (1,3), (2,2), (3,1), \dots\}.$$

So $N \times N$ is denumerable.

Note

$A \sim [0,1] \Leftrightarrow A$ has cardinality c and has the power of continuum.

13. Prove that the interval (a,b) has the power of the continuum.

Solution: Let a function $f : (0,1) \rightarrow (a,b)$ be

defined by $f(x) = a + (b-a)x$, for $x \in (0,1)$.

Take $x_1, x_2 \in (0,1)$

Let $f(x_1) = f(x_2)$

$$a + (b-a)x_1 = a + (b-a)x_2$$

$$(b-a)x_1 = (b-a)x_2$$

$$x_1 = x_2$$

∴ f is one-one.

Take any $y \in (a,b)$.

Choose $x = \frac{y-a}{b-a} \exists x \in (0,1)$ such that

$$f(x) = f\left(\frac{y-a}{b-a}\right)$$

$$= a + (b-a)\left(\frac{y-a}{b-a}\right)$$

$$= a + y - a = y$$

∴ f is onto.

Since f is one-one and onto, $(0,1) \sim (a,b)$.

Since $[0,1] \sim (0,1)$, so $[0,1] \sim (a,b)$.

But $\#([0,1]) = c$.

It follows that $\#((a,b)) = c$.

i.e. (a,b) has the power of the continuum.

14. Prove that the interval $[a,b]$ has the power of the continuum.

Solution: Let a function

$f : [0,1] \rightarrow [a,b]$ be defined by

$$f(x) = a + (b-a)x, \text{ for } x \in [0,1].$$

Take any $x_1, x_2 \in [0,1]$

Let $f(x_1) = f(x_2)$

$$a + (b-a)x_1 = a + (b-a)x_2$$

$$(b-a)x_1 = (b-a)x_2$$

$$x_1 = x_2$$

∴ f is one-one.

Take any $y \in [a,b]$.

Choose $x = \frac{y-a}{b-a} \exists x \in [0,1]$ such that

$$f(x) = f\left(\frac{y-a}{b-a}\right)$$

$$= a + (b-a)\left(\frac{y-a}{b-a}\right)$$

$$= a + y - a = y$$

∴ f is onto.

Since f is one-one and onto, $[0,1] \sim [a,b]$.

Since $[0,1] \sim [0,1]$, so $[0,1] \sim [a,b]$.

But $\#([0,1]) = c$.

It follows that $\#([a,b]) = c$.

i.e. $[a,b]$ has the power of the continuum.

15. Prove that the interval $(a,b]$ has the power of the continuum.

Solution: Let a function

$f : (0,1] \rightarrow (a,b]$ be defined by

$$f(x) = a + (b-a)x, \text{ for } x \in (0,1].$$

Take $x_1, x_2 \in (0,1]$

Let $f(x_1) = f(x_2)$

$$a + (b-a)x_1 = a + (b-a)x_2$$

$$(b-a)x_1 = (b-a)x_2$$

$$x_1 = x_2$$

∴ f is one-one.

Take any $y \in (a,b]$.

Choose $x = \frac{y-a}{b-a} \exists x \in (0,1]$ such that

$$f(x) = f\left(\frac{y-a}{b-a}\right) \\ = a + (b-a)\left(\frac{y-a}{b-a}\right) \\ = a + y - a = y$$

$\therefore f$ is onto.

Since f is one-one and onto, $(0,1] \sim (a,b]$.

Since $[0,1] \sim (0,1]$, so $[0,1] \sim (a,b]$.

But $\#([0,1]) = c$.

It follows that $\#((a,b]) = c$.

i.e. $(a,b]$ has the power of the continuum.

Cardinal Numbers

- (i) $\alpha = \#(A)$, $\beta = \#(B)$,
 (i) $\alpha + \beta = \#(A \cup B)$,
 (ii) $\alpha\beta = \#(A \times B)$.
 (2) $A \sim B \Rightarrow \#(A) = \#(B)$
 $A = B \Rightarrow \#(A) = \#(B)$

16. Let A_1, A_2, A_3 and A_4 be any sets. Define the sets B_1, B_2, B_3 and B_4 such that
 $\#(A_1) + \#(A_2) + \#(A_3) + \#(A_4) =$
 $\#(B_1 \cup B_2 \cup B_3 \cup B_4)$

Solution:

Let

$$B_1 = A_1 \times \{1\}$$

$$B_2 = A_2 \times \{2\}$$

$$B_3 = A_3 \times \{3\}$$

$$B_4 = A_4 \times \{4\}$$

Let the function $f: A_1 \rightarrow B_1$ be defined by
 $f(a) = (a, 1)$, $\forall a \in A_1$.
 $\therefore f$ is both one-one and onto.

$$\therefore A_1 \sim B_1$$

Hence $\#(A_1) = \#(B_1)$

Similarly, we can define $f_i: A_i \rightarrow B_i$ such that

$$f_i(a) = (a, i), a \in A_i, i = 1, 2, 3, 4.$$

Then f_i is both one-one and onto.

Hence $A_i \sim B_i, i = 1, 2, 3, 4$ and $B_i \cap B_j = \emptyset, i \neq j$.

$$\#(A_1) + \#(A_2) + \#(A_3) + \#(A_4) =$$

$$\#(B_1 \cup B_2 \cup B_3 \cup B_4)$$

17. Prove that $ac = c$.

Proof: $\mathbb{Z}$ = the set of integers

Let $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$, $A = [0, 1]$ and

$\mathbb{R}^\#$ = the set of real numbers.

Then $\#(Z) = a, \#(A) = c, \#(\mathbb{R}^\#) = c$.

Let $f: Z \times A \rightarrow \mathbb{R}^\#$ be defined by

$$f(i, a) = i + a, (i, a) \in Z \times A.$$

Let $f((i, a)) = f((i, b))$, $(i, a), (i, b) \in Z \times A$.

$$i + a = i + b$$

$$a = b$$

$$(i, a) = (i, b)$$

$\therefore f$ is one-one.

Let $y = i + a \in \mathbb{R}^\#$

Choose $x = (i, a)$.

$$\text{Then } \exists x \in Z \times A \text{ s.t. } f(x) = f((i, a)) \\ = i + a \\ = y.$$

$\therefore f$ is onto.

$$\therefore Z \times A \sim \mathbb{R}^\#$$

$$\#(Z \times A) = \#(\mathbb{R}^\#)$$

$$ac = c.$$

18. Prove that $\alpha^\beta \alpha^\gamma = \alpha^{\beta+\gamma}$, for any cardinal numbers α, β and γ .

Proof:

Let $\#(A) = \alpha, \#(B) = \beta, \#(C) = \gamma$,

where A, B, C are disjoint.

$$\#A^B = \#(A^B)$$

$$\#A^C = \#(A^C)$$

$$\alpha^B \alpha^C = \#(A^B \times A^C)$$

$$\beta + \gamma = \#(B \cup C)$$

$$\alpha^{\beta+\gamma} = \#(A)^{B \cup C}$$

Let $f: A^{B \cup C} \rightarrow A^B \times A^C$ be defined by

$F(g) = (g|_B, g|_C)$ where $g \in A^{B \cup C}$ and

$(g|_B, g|_C) \in A^B \times A^C$, $g|_B$ = the restriction of g to B

$g|_C$ = the restriction of g to C .

Clearly, F is one-one and onto.

$$A^{B \cup C} \sim A^B \times A^C$$

$$\therefore \alpha^{\beta+\gamma} = \alpha^\beta \cdot \alpha^\gamma.$$

R is a relation in N

R is an equivalence relation $\Leftrightarrow$

- (i) $x \in N \Rightarrow (x, x) \in R$ (R is reflexive)
- (ii) $x, y \in N, (x, y) \in R \Rightarrow (y, x) \in R$ (R is symmetric)
- (iii) $x, y, z \in N, (x, y) \in R, (y, z) \in R \Rightarrow (x, z) \in R$ (R is transitive)

19. Let R be the relation in the set of natural numbers N defined by " $x-y$ is divisible by 5". Prove that R is an equivalence relation.

Solution:

Let $x \in N$.

$\therefore (x - x) = 0$ is divisible by 5.

$\therefore (x, x) \in R$.

$\therefore R$ is reflexive.

Let $x, y \in N$ and $(x, y) \in R$.

$\therefore (x - y)$ is divisible by 5.

$$x - y = 5m, m \text{ is integer.}$$

$$y - x = -(x - y) = -5m$$

$\therefore y - x$ is divisible by 5.

$\therefore (y, x) \in R$

$\therefore R$ is symmetric.

Let $x, y, z \in N, (x, y) \in R$ and $(y, z) \in R$.

$\therefore (x, y)$ is divisible by 5 and

$(y - z)$ is divisible by 5.

$\therefore x - y = 5p$ and $y - z = 5q$, p, q are integers.

$$x - z = (x - y) + (y - z) = 5p + 5q = 5(p + q).$$

$(x - z)$ is divisible by 5.

$(x, z) \in R$.

R is transitive.

R is an equivalence relation.

20. Let α, β be cardinal numbers and $\alpha \leq \beta$.

Prove that for any cardinal number $\alpha^r \leq \beta^r$.

Proof:

Since $\alpha \leq \beta$, $\exists$ a set B with a subset A such that $\alpha = \#(A)$, $\beta = \#(B)$.

Let $r = \#(C)$.

$$\therefore \alpha^r = \#(A^C), \beta^r = \#(B^C)$$

(1) Let $f \in A^C$.

$$f: C \rightarrow A.$$

Since $A \subset B$, f can be considered as

$$f: C \rightarrow B.$$

$$f \in B^C$$

$$A^C \in B^C$$

$$A^C \prec B^C.$$

$A \subset B \Rightarrow A^C \subset B^C$

$$\#(A^c) \leq \#(B^c)$$

$$\therefore \alpha^r \leq \beta^r.$$

21. Prove that $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ for any cardinal number α, β and γ .

Proof:

Let $\alpha = \#(A)$, $\beta = \#(B)$ and $\gamma = \#(C)$ where A, B and C are pairwise disjoint sets

$$\begin{aligned} (\alpha + \beta) + \gamma &= \#(A \cup B) + \#(C) \\ &= \#((A \cup B) \cup C) \end{aligned}$$

$$\begin{aligned} \alpha + (\beta + \gamma) &= \#(A) + \#(B \cup C) \\ &= \#(A \cup (B \cup C)) \end{aligned}$$

$$\text{But } (A \cup B) \cup C = A \cup (B \cup C)$$

$$\#((A \cup B) \cup C) = \#(A \cup (B \cup C))$$

$$\therefore (\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Unit 2

Partially and Totally Ordered Sets

Partially Ordered Sets

A partial order in a set A is a relation R in A which is

- (i) reflexive, $(a, a) \in R, \forall a \in A$.
- (ii) anti-symmetric, $(a, b), (b, a) \in R \Rightarrow a = b$
- (iii) transitive, $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.

Totally Ordered Sets

If two elements in a partially ordered set A are comparable, then the partial order in A is called a total order in A .

Comparable

Two elements a, b of A are comparable if $a < b$ or $b < a$.

Eg2.1.3 Let X be any collection of sets and for all $x, y \in X$ we define $(x, y) \in R$ if $x \subseteq y$. Prove that R is a partial order in X .

Solution:

Let $x \in X$.

$x \subseteq x \Rightarrow (x, x) \in R$.

(i) $x \subseteq x$

$(x, x) \in R$

It is reflexive.

(ii) If $(x, y) \in R, (y, x) \in R$, then $x \subseteq y, y \subseteq x$.

$\therefore x = y$

It is anti-symmetric.

(iii) If $(x, y), (y, z) \in R$, then $x \subseteq y, y \subseteq z$.

$x \subseteq z$

$(x, z) \in R$

It is transitive.

R is a partial order relation.

Eg2.1.4 Let R be a relation in N defined by " x is a multiple of y ". Prove that R is a partial order in N .

Proof

Let $N = \{1, 2, 3, \dots\}$

" x is a multiple of y " $\Rightarrow (x, y) \in R$

(i) Since $x = 1 \cdot x$, x is a multiple of x

$(x, x) \in R$

It is reflexive.

(ii) If $(x, y) \in R$, x is a multiple of y .

$x = my$ — (1)

If $(y, x) \in R$, y is a multiple of x .

$y = nx$ — (2)

(1) $\Rightarrow x = m(ny)$

$= (mn)x$

$mn = 1$

$m = 1, n = 1 (\because m, n \in \mathbb{Z})$

$\therefore x = y$.

If $(x, y) \in R, (y, x) \in R \Rightarrow x = y$.

It is anti-symmetric.

(iii) If $(x, y), (y, z) \in R$, then

" x is a multiple of y " and

" y is a multiple of z "

$x = my$ — (3), $y = nz$ — (4)

(3) $\Rightarrow x = m(nz)$

$= (mn)z$

$\therefore x$ is a multiple of z .

$\therefore (x, z) \in R$

$(x, y), (y, z) \in R \Rightarrow (x, z) \in R$

It is transitive.

$\therefore R$ is a partial order relation in N .

Exm / Pgs
 1. Let R be the relation in N , the set of natural numbers, defined by " x divides y ". Prove that (N, R) is a partially ordered set.

Proof:

Let $N = \{1, 2, 3, \dots\}$

" x divides y " $\Rightarrow (x, y) \in R$

(i) x divides x

$$\therefore (x, x) \in R$$

It is reflexive.

(ii) If $(x, y) \in R$, then x divides y .

$$y = mx \quad (1)$$

If $(y, x) \in R$, then y divides x .

$$x = ny \quad (2)$$

$$(1) \Rightarrow y = mx = m(ny) = (mn)y.$$

$$\therefore mn = 1$$

$$m = 1, n = 1$$

$$\therefore x = y$$

$$\text{If } (x, y) \in R, (y, x) \in R \Rightarrow x = y$$

It is anti-symmetric.

(iii) If $(x, y), (y, z) \in R$, then

x divides y and y divides z

$$y = mx \quad (3)$$

$$z = ny \quad (4)$$

$$(4) \Rightarrow z = n(mx) = (n, m)x$$

$\therefore x$ divides z .

$$(x, z) \in R$$

$$(x, y), (y, z) \in R \Rightarrow (x, z) \in R$$

It is transitive.

$\therefore R$ is a partial order relation in N .

Exm
 2. Let $X = \{1, 2, 3\}$ and $\mathcal{P}(X)$ be the power set of X . Let R be the relation in $\mathcal{P}(X)$ defined by " A is a subset of B ". Prove that R is a partial order in $\mathcal{P}(X)$

Proof:

Let $X = \{1, 2, 3\}$ and

$\mathcal{P}(X)$ = the power of X

$$= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

A is a subset of $B \Rightarrow (A, B) \in R$.

(i) A is a subset of A

$$A \subset A \Rightarrow (A, A) \in R$$

It is reflexive.

(ii) If $(A, B) \in R$, then A is a subset of B .

$$A \subset B \quad (1)$$

If $(B, A) \in R$, then B is a subset of A .

$$B \subset A \quad (2)$$

By (1) and (2), $A = B$

It is anti-symmetric.

(iii) If $(A, B), (B, C) \in R$, then

A is a subset of B and B is a subset of C .

$$A \subset B, B \subset C$$

$$\therefore A \subset C$$

$$(A, C) \in R$$

It is transitive.

$\therefore R$ is a partial order relation.

3. Determine whether or not each of the following sets is totally ordered.

(1) $X = \{2, 6, 24\}$, the relation R in X is defined by " x divides y ". *Exm yaq amon*

(2) $X = \{2, 8, 32, 4\}$, the relation R in X is defined by " x divides y ".

(3) $X = \{3, 15, 5\}$, the relation R in X is defined by " x divides y ".

(4) $X = \{7\}$, the relation R in X is defined by " x divides y ".

Solution:

$\mathcal{P}$

$\mathcal{P}$

$$X = \{1, 2, 3\}$$

$$= \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

- (1) 24 2 divides 6 and 6 divides 24.

↑ $\therefore 2$ divides 24.

6 ~~$2 < 6 < 24$~~ $2 < 6 < 24$

↑ $\{2, 6, 24\}$ is totally ordered set.

2

- (2) 32 2 divides 4, 4 divides 8 and 8 divides 32.

↑ $\therefore 2$ divides 32.

8 ~~$2 < 4 < 8 < 32$~~ $2 < 4 < 8 < 32$

↑ $\{2, 4, 8, 32\}$ is totally ordered set.

4

↑

2

- (3) 15



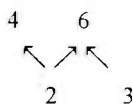
Since 3 and 5 are not comparable,
 $\{3, 5, 15\}$ is not totally ordered set.

- (4) Since 7 divides 7, $\{7\}$ is totally ordered set.

4. Let $V = \{2, 3, 4, 6\}$. Prove that V is a partially ordered set under divisibility. Construct a diagram of (V, R) . Here $(a, b) \in R$ means a divides b .

Solution:

Let $V = \{2, 3, 4, 6\}$



- (i) Every $a \in V$ divides a itself.

$$(a, a) \in R$$

R is reflexive.

- (ii) If $(x, y) \in R, (y, x) \in R$, then
 x divides y and y divides x .

$$y = mx \rightarrow (1)$$

$$x = ny \rightarrow (2)$$

$$y = mx = m(ny) = (mn)y$$

$$mn = 1$$

$$\therefore m = 1, n = 1$$

$$x = y$$

R is anti-symmetric.

- (iii) If $(x, y) \in R, (y, z) \in R$, then

x divides y and y divides z .

$$y = mx \text{ and } z = ny.$$

$\therefore x$ divides z .

It is transitive.

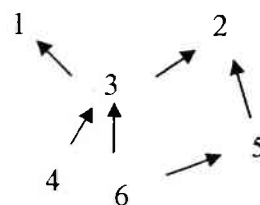
$\therefore V$ is a partial order relation.

$$z = n(m)x$$

$$z = nmnx$$

$$z = r \cdot y$$

6. Let $W = \{1, 2, 3, 4, 5, 6\}$ be ordered as follows:

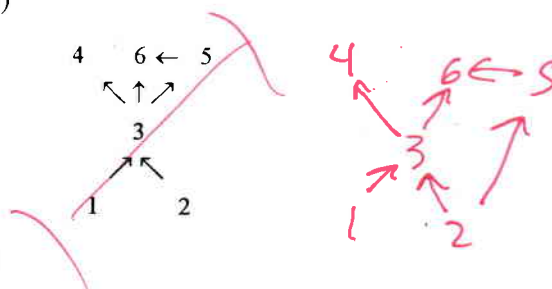


- (a) Construct the diagram of the element in which defines the inverse order.

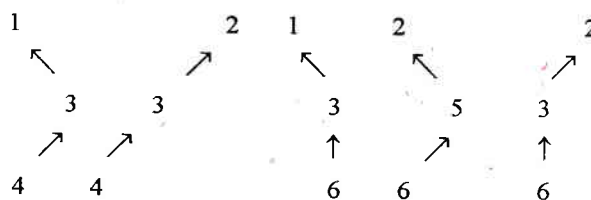
- (b) Find all the totally ordered subsets of W , each of which at least three elements.

Solution:

- (a)



- (b)



$\{1, 3, 4\}, \{2, 3, 4\}, \{1, 3, 6\}, \{2, 5, 6\}, \{2, 3, 6\}$ are all totally ordered subsets of W .

7. Let $\mathcal{A}$ be the family of all subsets of A of N , the set of natural numbers, where A has the following properties: *a c p*

A is finite and the greatest common divisor of the elements of A is 1. *1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

State whether or not each of the following subsets of N belongs to $\mathcal{A}$

- (a) $\{2, 3, 8\}$, (b) $\{4, 6, 8\}$, (c) $\{2, 3, 5, 8\}$, (d) $\{2, 5\}$.

Solution:

(a) $\{2, 3, 8\} \in \mathcal{A}$ since the greatest common divisor of $\{2, 3, 8\}$ is 1. *2 = 2x1, 3 = 3x1, 8 = 2x2x2, GCD = 1*

(b) $\{4, 6, 8\} \notin \mathcal{A}$ since the greatest common divisor of $\{4, 6, 8\}$ is 2.

(c) $\{2, 3, 5, 8\} \in \mathcal{A}$ since the greatest common divisor of $\{2, 3, 5, 8\}$ is 1.

(d) $\{2, 5\} \in \mathcal{A}$ since the greatest common divisor of $\{2, 5\}$ is 1.

First and Last Elements

Let A be ordered set.

$a \in A$ is first element if $a \leq x, \forall x \in A$.

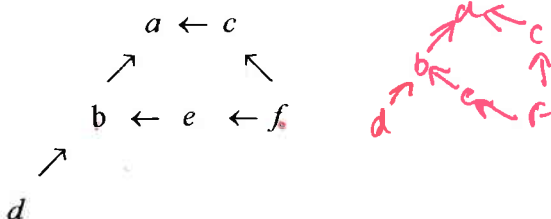
$a \in A$ is last element if $x \leq a, \forall x \in A$.

Maximal and Minimal Elements

$a \in A$ is minimal element if $a \leq x \Rightarrow a = x$.

$a \in A$ is maximal element if $x \leq a \Rightarrow a = x$.

8. Let $B = \{a, b, c, d, e, f\}$ be ordered as follows:



(a) Find all the minimal elements of B .

Ans: d, b, e are minimal elements

(b) Find all the maximal elements of B .

(c) Does B have a first elements?

(d) Does B have a last elements?

Solution:

(a) minimal elements of $B = d, b, e$.

(b) maximal elements of $B = a, f$.

(c) B does not have a first elements.

(d) Last element = a .

9. Let $B = \{2, 3, 4, 5, 6, 8, 9, 10\}$ be ordered by

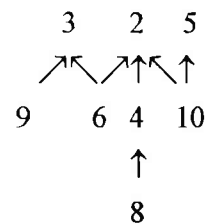
" x is a multiple of y ". Construct a diagram of B .

(a) Find all the minimal elements of B .

(b) Find all the maximal elements of B .

(c) Does B have a first or a last element?

Solution:



(a) Maximal elements = $3, 2, 5$.

(b) Minimal elements = $9, 6, 8, 10$.

(c) B does not have a first or a last element.

10. Let A be a totally ordered set and a and b be minimal elements in A . Prove that $a = b$.

Solution:

Since A is totally ordered set, the elements a and b are comparable.

$\therefore a \leq b$ (or) $b \leq a$. *a < b (or) b < a*

Since b is minimal element, $a \leq b \Rightarrow a = b$.

Since a is minimal element, $b \leq a \Rightarrow a = b$.

In either case, $a = b$.

Upper and Lower Bounds

Let B be a subset of a partially ordered set A .

An element $m \in A$ is called an upper bound of

B if $x \leq m, x \in B$.

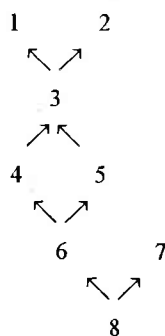
An element $m \in A$ is called a lower bound of

B if $m \leq x, x \in B$.

$\sup(B)$ = least upper bound of B

$\inf(B)$ = greatest lower bound of B

11. Let $W = \{1, 2, \dots, 7, 8\}$ be ordered as follows:



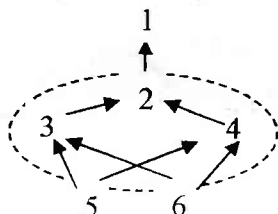
Consider the subset $V = \{4, 5, 6\}$ of W .

- Find the set of upper bounds of V .
- Find the set of lower bounds of V .
- Does $\sup(V)$ exist?
- Does $\inf(V)$ exist?

Solution:

- the set of upper bounds of $V = \{1, 2, 3\}$.
- the set of lower bounds of $V = \{6, 8\}$.
- $\sup(V) = 3$.
- $\inf(V) = 6$.

12. Let $D = \{1, 2, 3, 4, 5, 6\}$ be ordered as follows:



Consider the subset $E = \{2, 3, 4\}$ of D .

- Find the set of upper bounds of E .
- Find the set of lower bounds of E .
- Does $\sup(E)$ exist?
- Does $\inf(E)$ exist?

Solution:

- the set of upper bounds of $E = \{1, 2\}$.
- the set of lower bounds of $E = \{5, 6\}$.
- $\sup(E) = 2$.
- $\inf(E)$ does not exist.

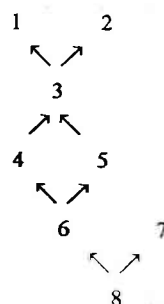
13. Consider $\mathbb{Q}$, the set of rational numbers and its subset $A = \{x \mid x \in \mathbb{Q}, x^3 < 3\}$.

- Is A bounded above?
- Is A bounded below?
- Does $\sup(A)$ exist?
- Does $\inf(A)$ exist?

Solution:

- A is bounded above.
- A is not bounded below.
- $\sup(A)$ does not exist.
- $\inf(A)$ does not exist.

14. Let $W = \{1, 2, \dots, 7, 8\}$ be ordered as follows:



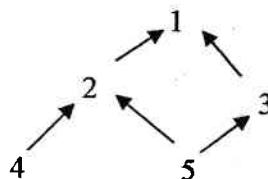
Consider the subset $A = \{4, 5, 7\}$ of W .

- Find the set of upper bounds of A .
- Find the set of lower bounds of A .
- Does $\sup(A)$ exist?
- Does $\inf(A)$ exist?

Solution:

- the set of upper bounds of $A = \{1, 2, 3\}$.
- the set of lower bounds of $A = \{8\}$.
- $\sup(A) = 3$.
- $\inf(A) = 8$.

15. Let $B = \{1, 2, 3, 4, 5\}$ be ordered as follows:

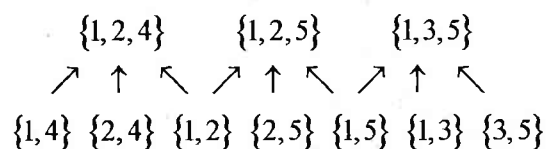


Let $\mathcal{B}$ be the family of all totally ordered subsets of B which contains 2 or more elements and let B be partially ordered by set inclusion. Construct a diagram of $\mathcal{B}$.

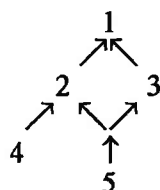
Solution:

The totally ordered subsets of

$$B = \{1, 2, 4\}, \{1, 2, 5\}, \{1, 3, 5\}, \{1, 2\}, \{1, 4\}, \\ \{2, 4\}, \{1, 5\}, \{2, 5\}, \{1, 3\}, \{3, 5\}.$$



16. Let $B = \{1, 2, 3, 4, 5\}$ be ordered as follows:



- Find all the minimal elements of B .
- Find all the maximal elements of B .
- Does B have a first elements?
- Does B have a last elements?

Solution:

- minimal elements of $B = 4, 5$.
- maximal elements of $B = 1$.
- B does not have a first elements.
- Last element $= 1$.

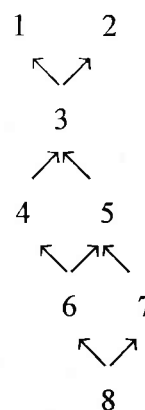
17. Consider Q , the set of rational numbers and its subset $A = \{x \in Q, 8 < x^3 < 15\}$.

- Is A bounded above?
- Is A bounded below?
- Does $\sup(A)$ exist?
- Does $\inf(A)$ exist?

Solution:

- A is bounded above.
- A is bounded below.
- $\sup(A)$ does not exist.
- $\inf(A)$ exist and $\inf(A) = 2$.

18. Let $W = \{1, 2, \dots, 7, 8\}$ be ordered as follows:



(1) Consider the subset $A = \{2, 3, 6\}$ of W .

- Find the set of upper bounds of A .
- Find the set of lower bounds of A .
- Does $\sup(A)$ exist?
- Does $\inf(A)$ exist?

Solution:

- the set of upper bounds of $A = \{2\}$.
- the set of lower bounds of $A = \{6, 8\}$.
- $\sup(A) = 2$.
- $\inf(A) = 6$.

(2) Consider the subset $B = \{1, 2, 4, 7\}$ of W .

- Find the set of upper bounds of B .
- Find the set of lower bounds of B .
- Does $\sup(B)$ exist?
- Does $\inf(B)$ exist?

Solution:

- the set of upper bounds of $B = \emptyset$.
- the set of lower bounds of $B = \{2\}$.
- $\sup(B)$ does not exist.
- $\inf(B)$ exist and $\inf(B) = 8$.

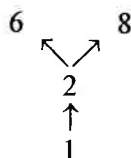
Similar Sets

An order set A is similar to an ordered set B , denoted by $A \approx B$, if there exists a function $f: A \rightarrow B$ which is one-one and onto and which has the property that, for any elements $a, b \in A$, $a < b \Leftrightarrow f(a) < f(b)$. The function is called a similarity mapping of A into B .

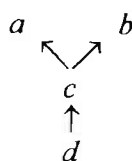
19. Let $V = \{1, 2, 6, 8\}$ be ordered by “ x divides y ” and $W = \{a, b, c, d\}$ be ordered by the following diagram. Show that $V \approx W$.

Solution:

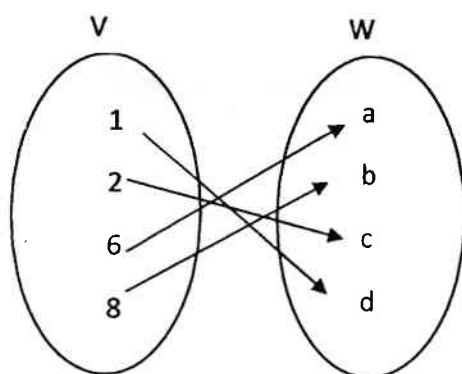
$$V = \{1, 2, 6, 8\}$$



$$W = \{a, b, c, d\}$$



Let $f: V \rightarrow W$ be defined by



Clearly, f is one-one and onto.

Take any $p, q \in V$.

$$p < q \Leftrightarrow f(p) < f(q)$$

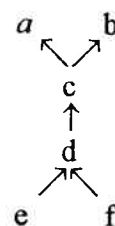
Therefore f is a similarity mapping of V into W .

Hence $V \approx W$.

20. Give an example of an ordered set $X = (A, R)$ which is similar to $Y = (A, R^{-1})$, the set A with the inverse order.

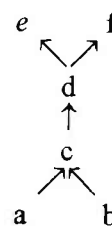
Solution:

Consider $W = \{a, b, c, d, e, f\}$ ordered as follows:



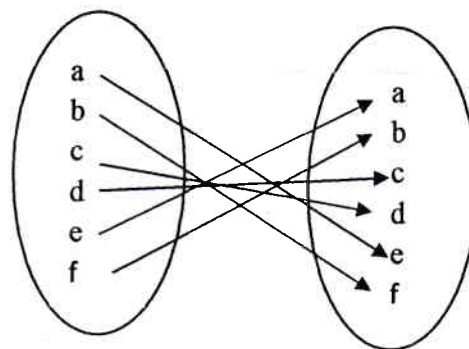
$$X = (A, R)$$

The inverse order of W as follows:



$$Y = (A, R^{-1})$$

Let $f: X \rightarrow Y$ be defined by



Clearly, f is one-one and onto.

Take any $p, q \in X$.

$$p < q \Leftrightarrow f(p) < f(q)$$

Therefore f is a similarity mapping of X into Y .

Hence $X \approx Y$.